



Wolf Analyzer



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Tools We Used

01

Matplotlib

02

Pandas

03

Numpy

04

Seaborn

05

Plotly

06

Streamlit

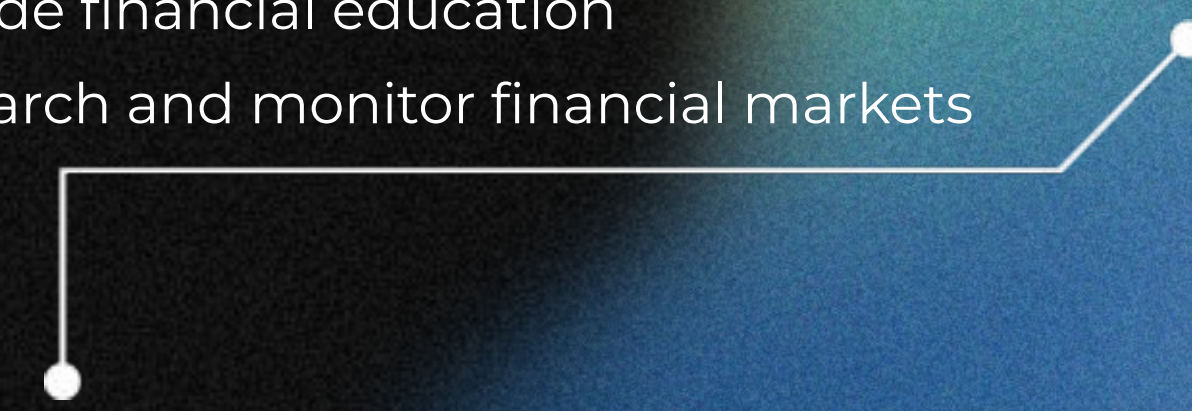
07

Scikit-Learn

08

SQL

Overview

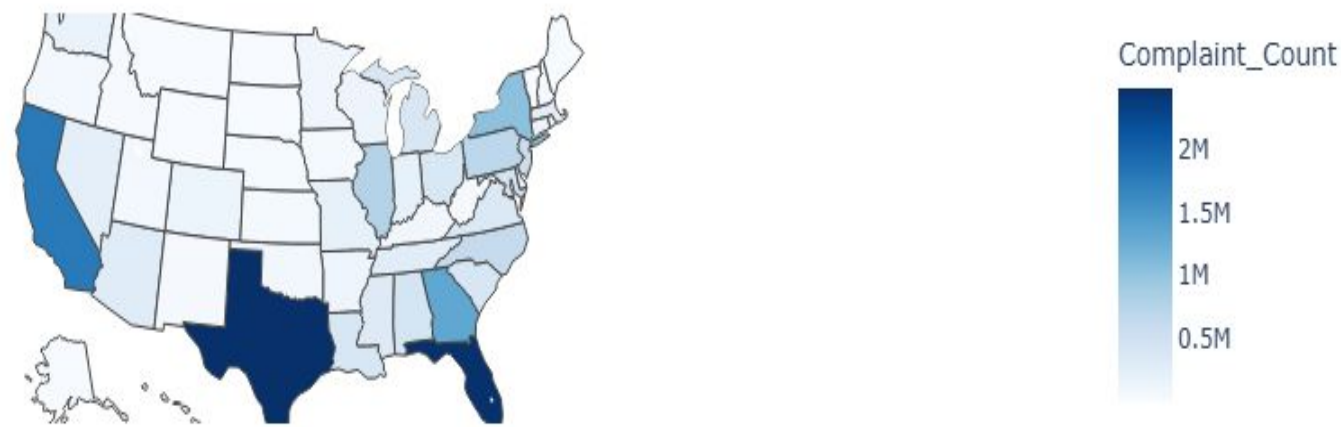
- The Consumer Financial Protection Bureau (CFPB) works to create and support innovative and resilient consumer financial markets where consumers can choose the products and services that meet their individual needs.
 - Main functions of the CFPB:
 - Enforce consumer-financial laws
 - Handle consumer complaints
 - Protect consumers from unfair practices
 - Combat discrimination in consumer finance
 - Provide financial education
 - Research and monitor financial markets
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Questions

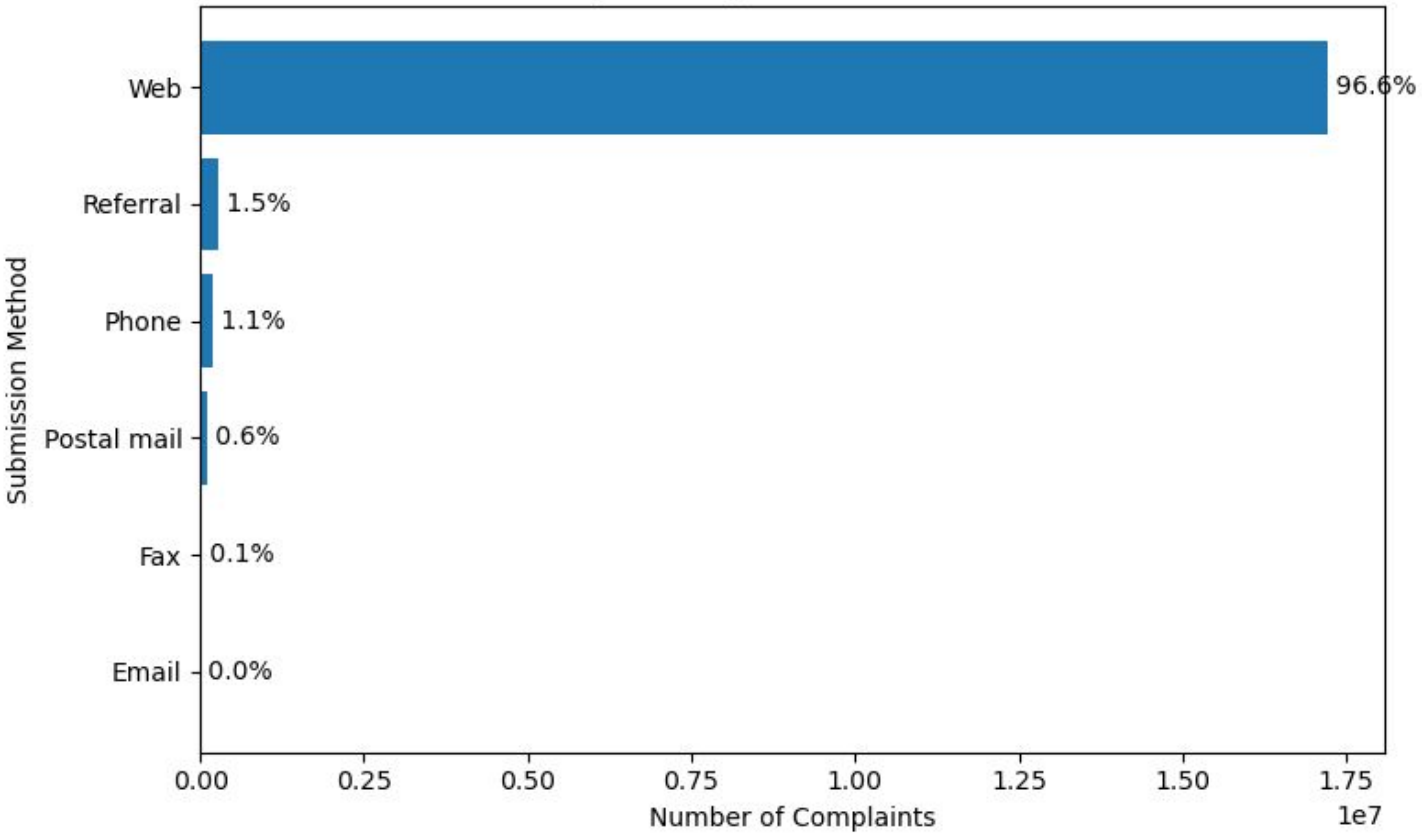
1. How many complaints originate per state?
2. In what way do customers tend to submit their complaints the most?
3. What does the number of complaints look over the years?
4. How many complaints are there for each product?
5. What was the most frequent company's response to the consumers?
6. What product received the most complaints?



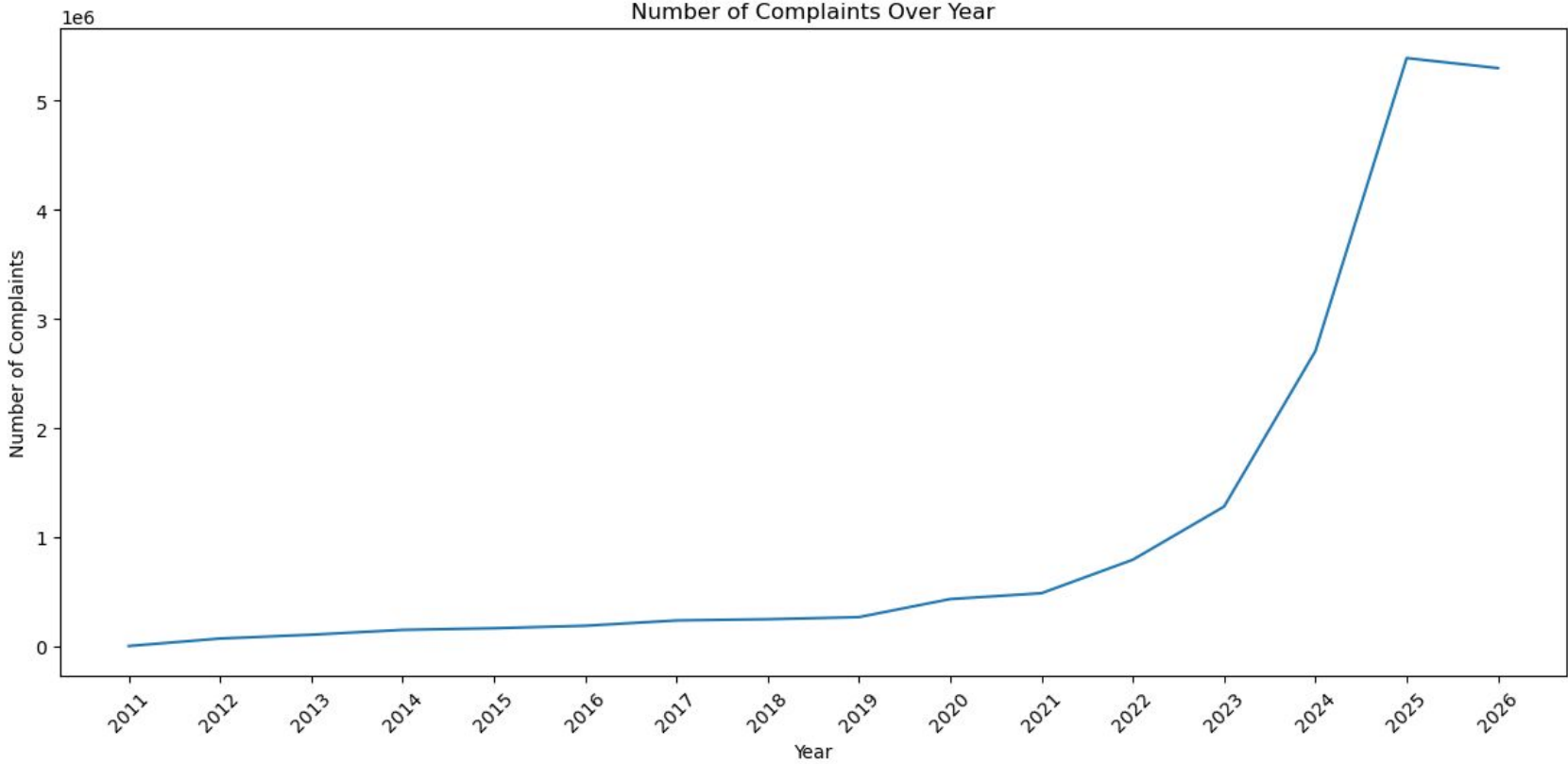
Number of Complaints by State



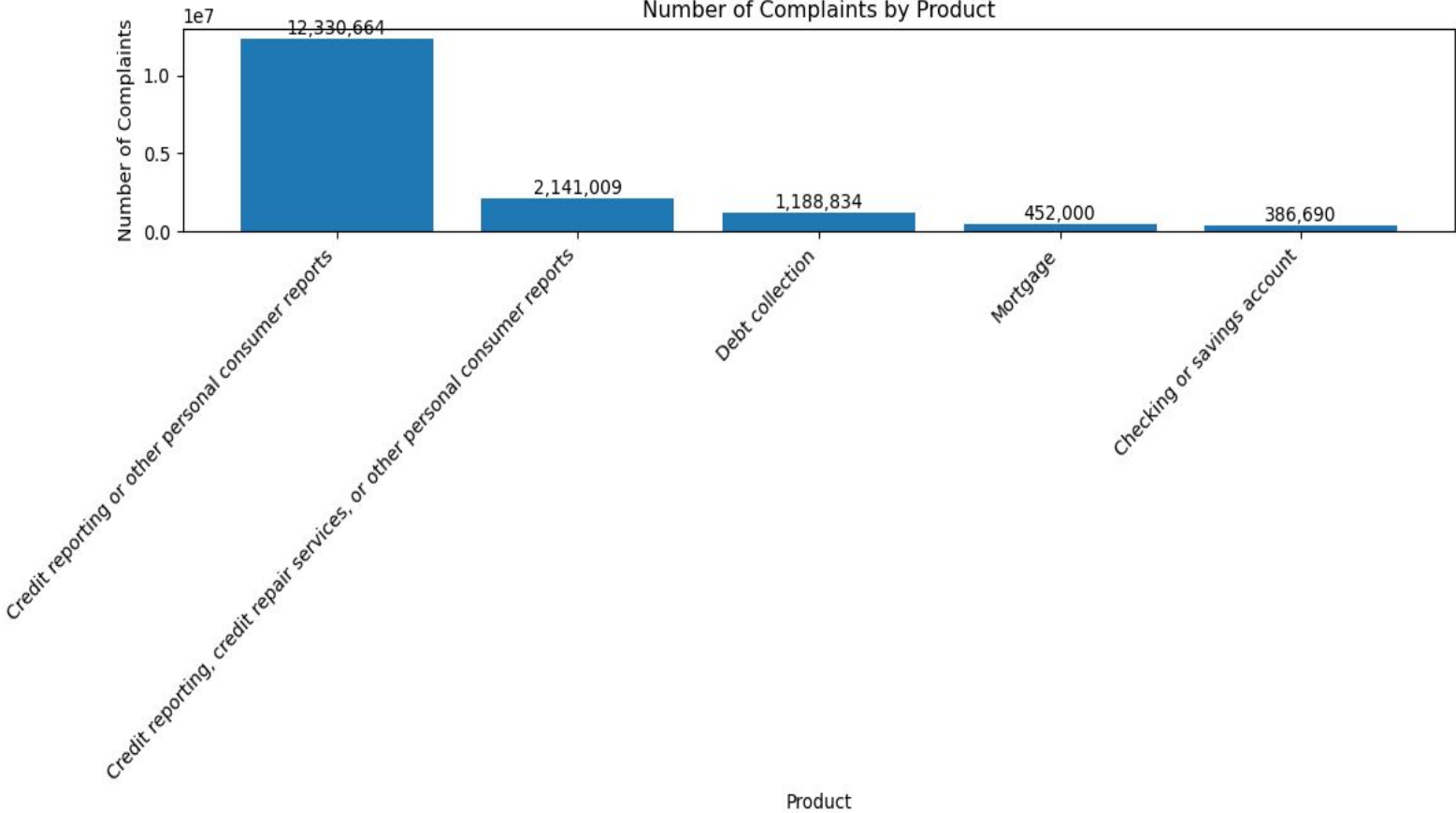
Complaints by Submission Method



Number of Complaints Over Year



Number of Complaints by Product

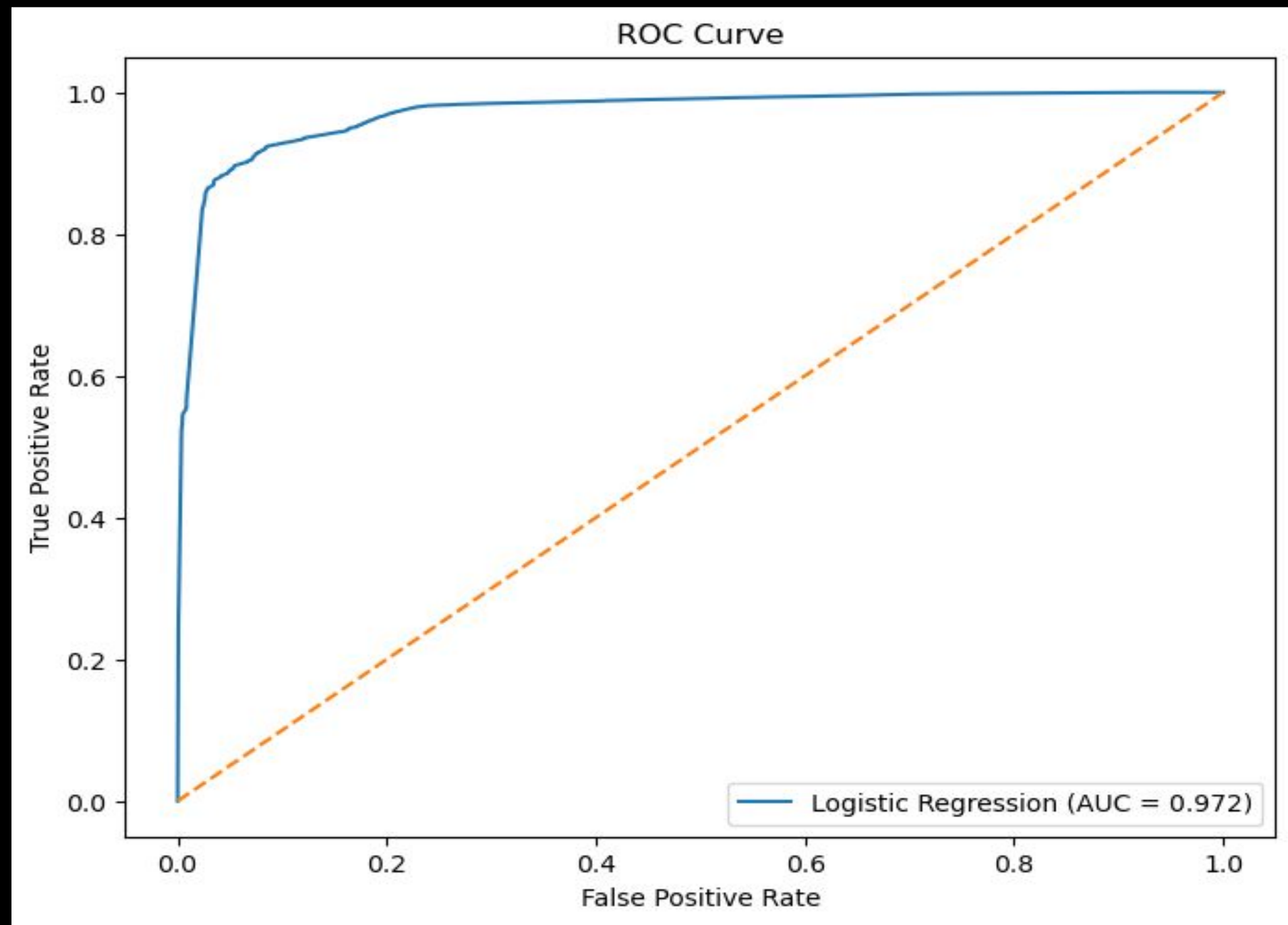
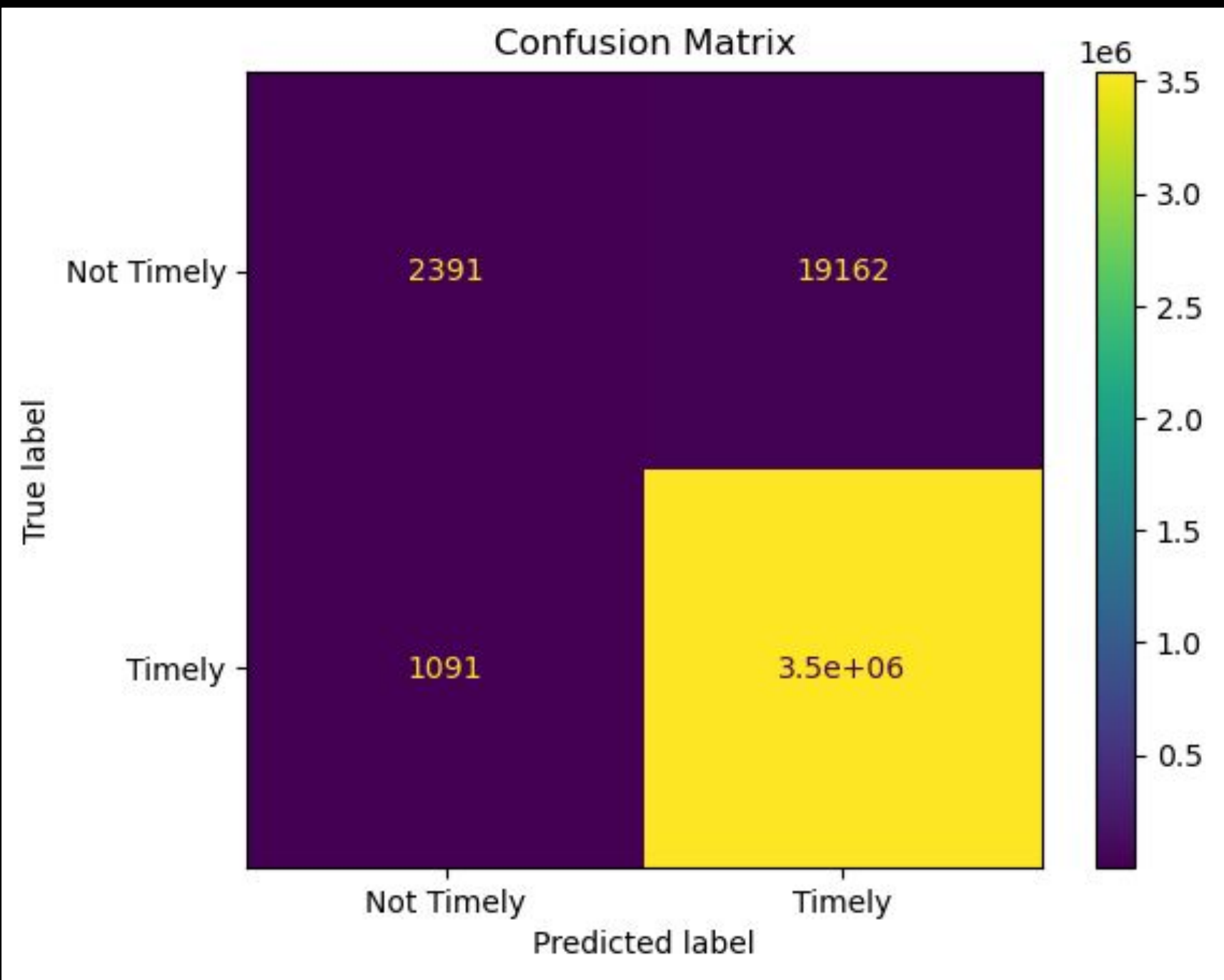


	Company response to consumer	number_of_responses
1	Closed with explanation	10924934
3	Closed with non-monetary relief	5982192
6	In progress	625000
2	Closed with monetary relief	214730
8	Untimely response	33455
5	Closed without relief	17625
0	Closed	17272
4	Closed with relief	5213
7	Unknown	21

	Product	number_of_products
6	Credit reporting or other personal consumer re...	12330664
7	Credit reporting, credit repair services, or o...	2141009
8	Debt collection	1188834
12	Mortgage	452000
1	Checking or savings account	386690
3	Credit card	341480
4	Credit card or prepaid card	199806
10	Money transfer, virtual currency, or money ser...	181274
5	Credit reporting	138413
18	Student loan	131143
19	Vehicle loan or lease	101832
0	Bank account or service	83575
16	Payday loan, title loan, personal loan, or adv...	38052
2	Consumer Loan	31082
15	Payday loan, title loan, or personal loan	29719
17	Prepaid card	22991
9	Debt or credit management	10291
14	Payday loan	5464
11	Money transfers	5087
13	Other financial service	1019
20	Virtual currency	17

Logistic Regression





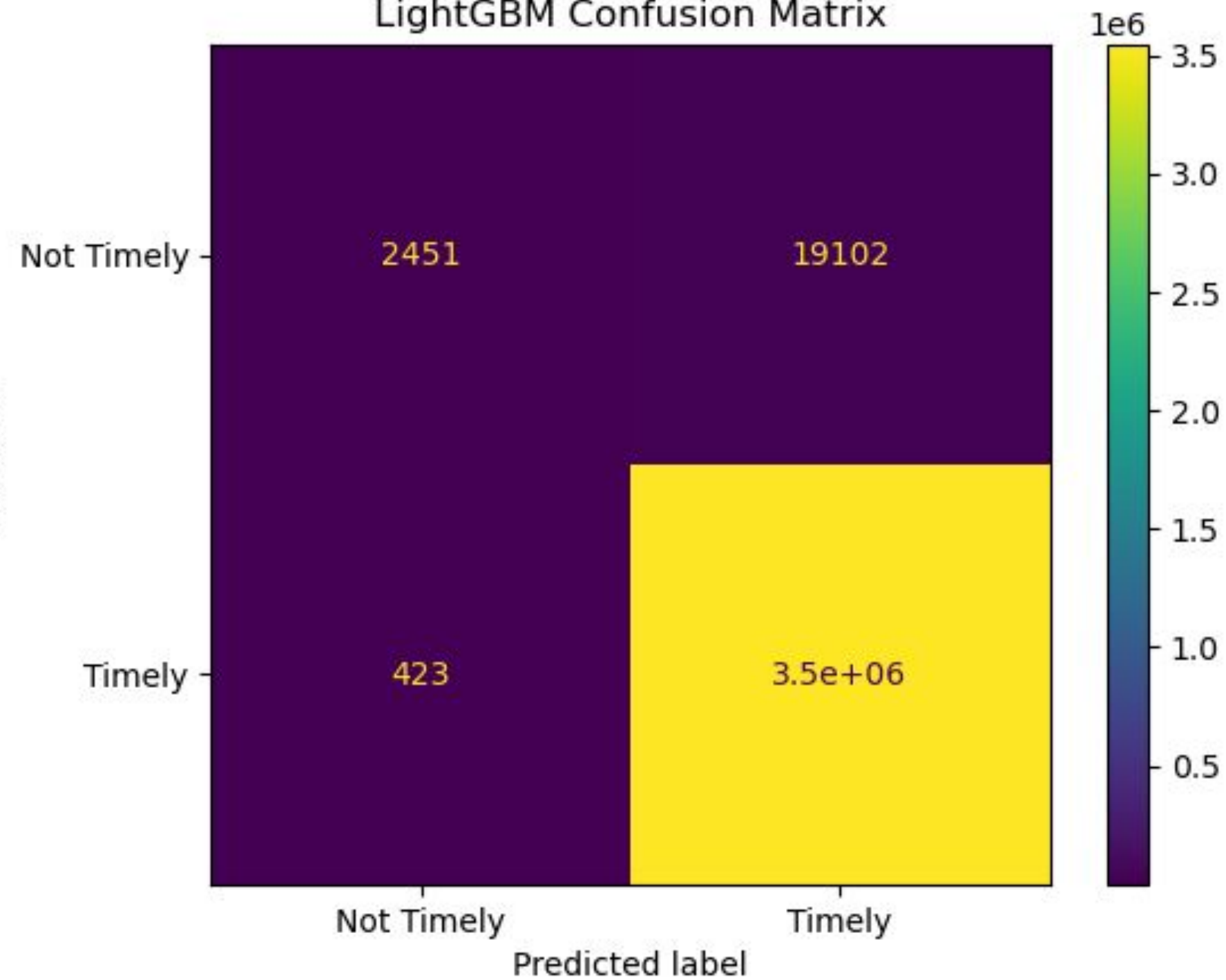
Test accuracy: 0.9943174819708487

	precision	recall	f1-score	support
0	0.69	0.11	0.19	21553
1	0.99	1.00	1.00	3542536
accuracy			0.99	3564089
macro avg	0.84	0.56	0.59	3564089
weighted avg	0.99	0.99	0.99	3564089

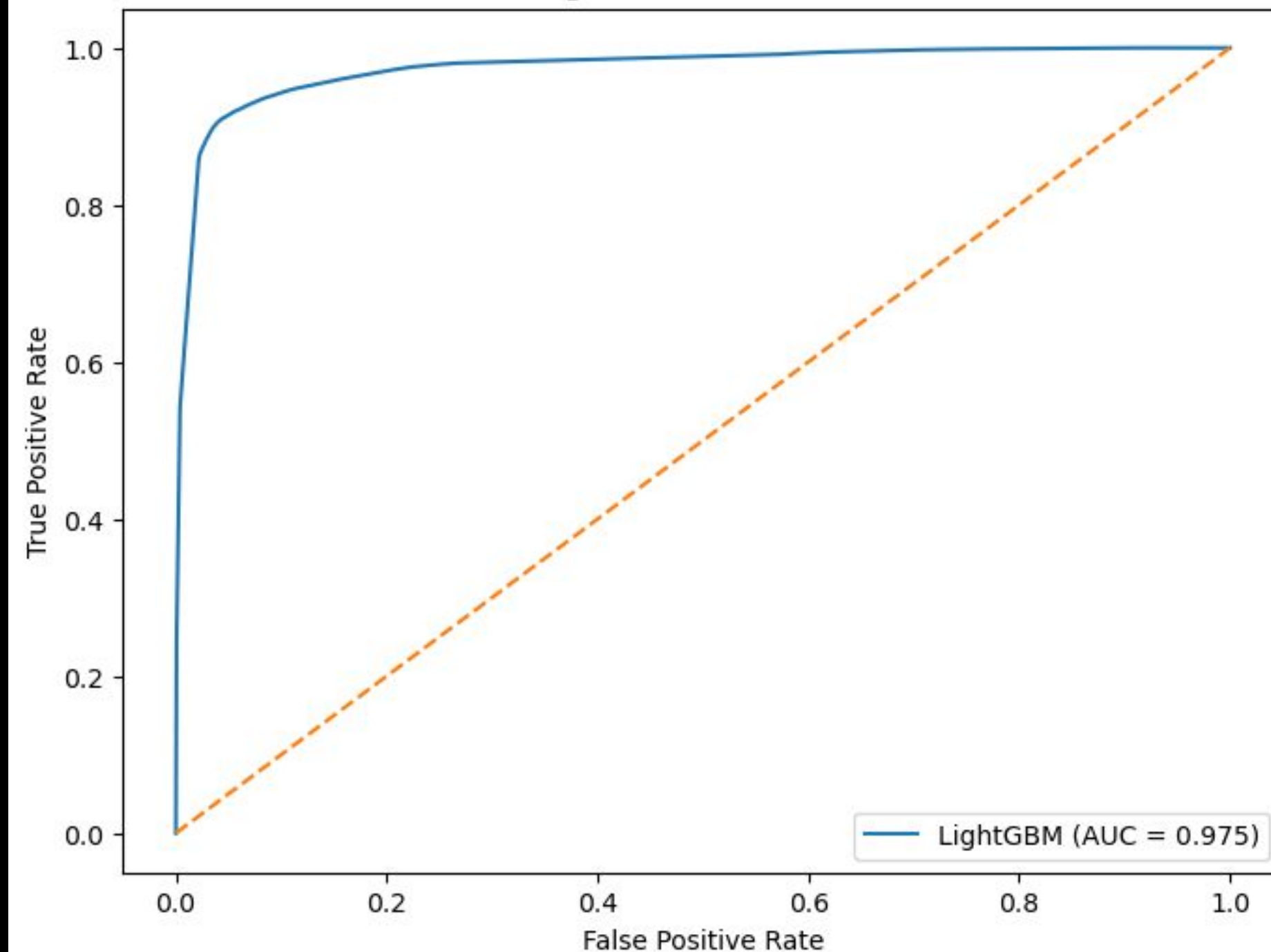
LightGBM



LightGBM Confusion Matrix



LightGBM ROC Curve



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Accuracy: 0.9945
Precision: 0.9946
Recall: 0.9999
F1 Score: 0.9973
ROC-AUC: 0.9748
    
```




Our Website

A stylized illustration of a laptop with a white outline. The screen area is a solid black rectangle containing the text 'Our Website' in a bold, white, sans-serif font. The laptop has a thin white base. To the left of the laptop, there are three white line segments with circular endpoints, resembling circuit traces or data paths. One line extends horizontally from the left edge, then angles up and right to a dot. Another line extends horizontally from the left edge, then angles down and right to a dot. A third line extends horizontally from the left edge, then angles up and right to a dot. To the right of the laptop, a single white line segment extends horizontally from the right edge of the screen, then angles up and right to a dot.



Thank You!